

The Structural Analysis of the Static Fire Test Stand

Introduction

The development of rocket engines requires thorough testing and data collection. A static fire is one of the most important tests used for iterative development of the engine. It is conducted by restricting the engine in all its degrees of freedom and firing it to collect data on the inlet pressures and impulse. These requirements require the design of a robust test-stand that can support the development of rocket engines throughout the years. This report will be dedicated to conduct structural analysis of the test stand at Waterloo Rocketry by conducting Finite Element Analysis (FEA) using ANSYS.

FEA is the process of predicting an object's behavior using the Finite Element Method (FEM).

[1] The FEM uses mathematics to split an object into many smaller elements. The properties of these elements are then calculated individually to understand the behaviour of complex systems more accurately. The application of the FEM requires information on its background, applications, and methodologies. To conduct FEA, an understanding of its methodology is required: the following outlines a general procedure.

- 1) Problem classification
- 2) Defining material properties.
- 3) Geometry creation
- 4) Model definition.
- 5) Model idealization.
- 6) Meshing and mesh refinement.
- 7) Specify initial conditions and limitations.
- 8) Loads and boundary conditions.
- 9) Methods of solution

Geometry Creation and Idealization

The model was originally imported from SolidWorks as a Parasolid and refined using SpaceClaim. Figure 1 shows the test stand assembly in SolidWorks.



Figure 1: SolidWorks Model of the Test Stand

The assembly was simplified to include the tubes and the bolts to reduce the complexity of the problem. The tube assembly is the main component that distributes the forces across the system therefore only this component was considered. The bolts and brackets were included to model pin joint behaviour in the simulation.

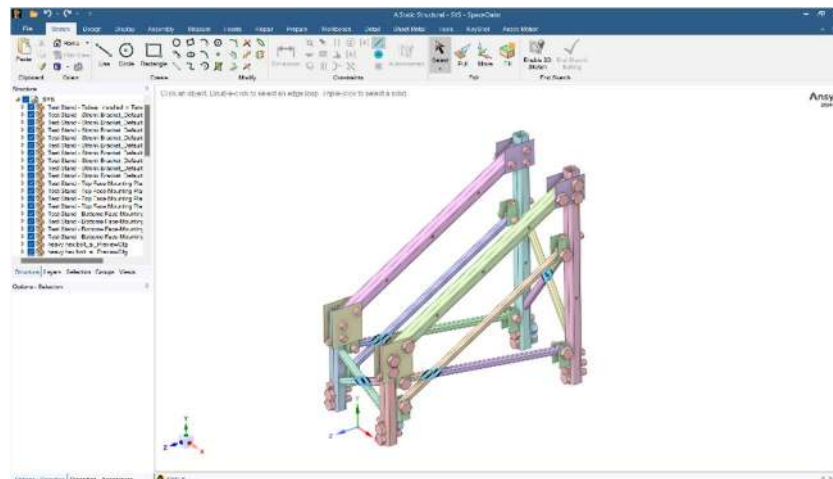


Figure 2: Final Geometry of the Test Stand in Space Claim

Material Data

Material behaviour governs how the beams respond to stress as it is distributed throughout the structure. The tube structure and the brackets are made from cold rolled AISI 1020 steel [2]. The bolts are made from Grade 5 Zinc-plated 304 stainless steel [3] .

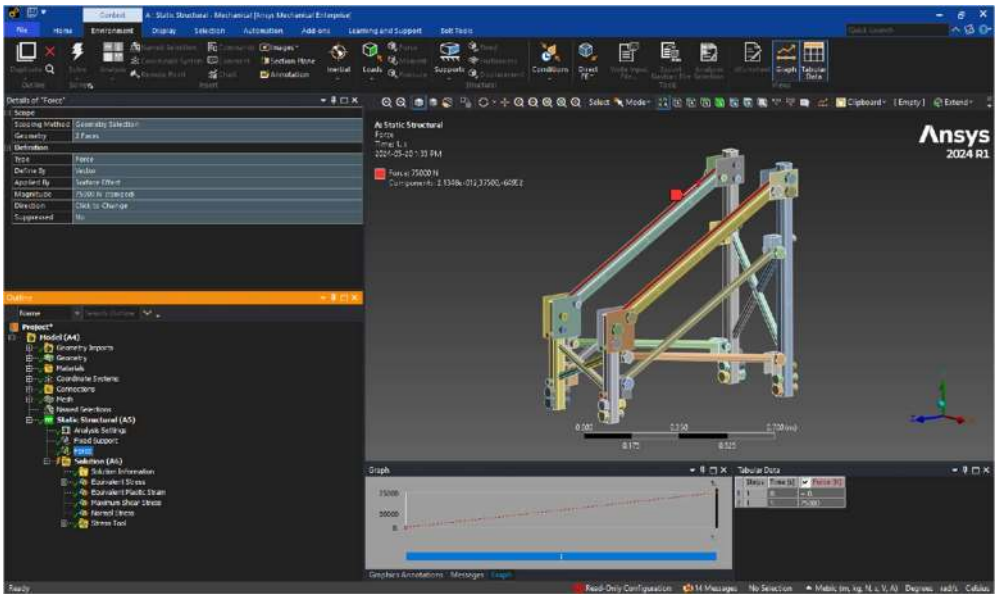
Table 1: Material Properties

	AISI 1020 Steel, Cold rolled [2]	304 Stainless Steel [4]
Young’s Modulus	186 [GPa]	193 [GPa]
Poisson’s Ratio	0.29	0.29
Density	7.87 [g/cc]	8 [g/cc]
Yield Strength	350 [MPa]	215 [MPa]
Ultimate Tensile Strength	420[MPa]	505 [MPa]

The test stand was also made using plain carbon steel, AISI 1020 is most commonly used Carbon steel for structural applications, the It is important to note that the test stand was painted with corrosion resistive paint after it was assembled. Corrosion was therefore not considered when considering the condition of the test stand.

Loads and constraints

The force is applied in the direction that the thrust from the engine would transfer onto the test



stand the force is symmetrically applied to both sides of the test stand. A steady force of 10 000N is applied, with increments until the test stand reached yield.

Figure 3: Configuration of Loads on System

The test stand is constrained by creating fixed supports on the bottom surfaces of the legs, circled in red as seen in figure 4. This simulated the test stand being bolted to the ground at the test site.

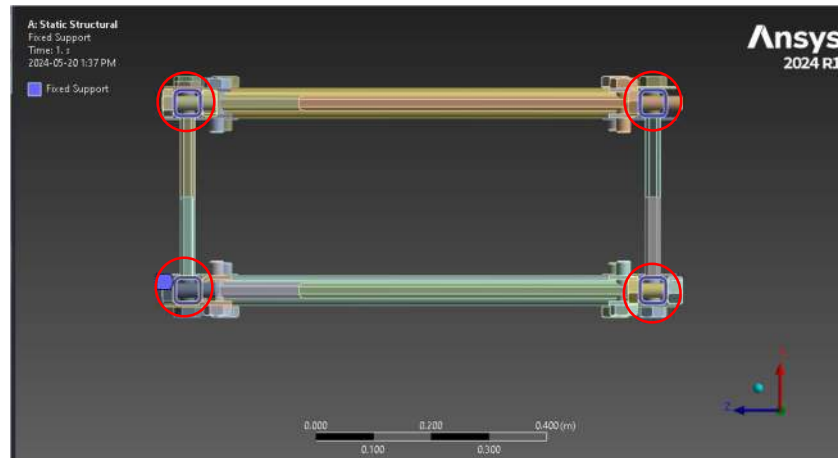


Figure 4: Fixed Support Locations

Joints

The joints of the assembly were modelled using the connections tool in Ansys Mechanical. The connections of the support tubes were modelled as revolute joints, modelling pin joint behaviour as shown in Figure 5.

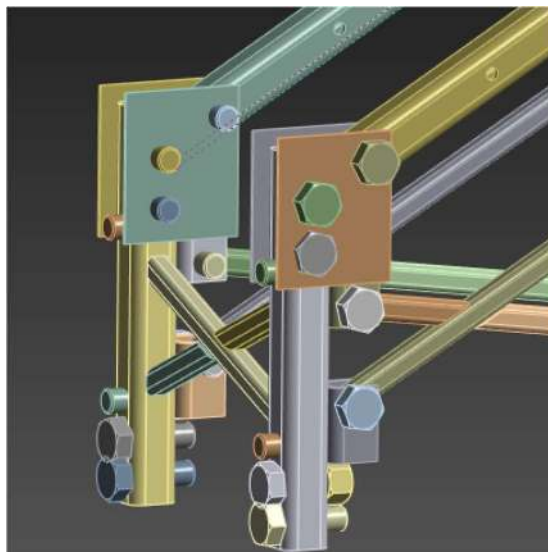


Figure 5: Pin Joint Behavior

Welded joints were modelled using the weld meshing tool. The contact surfaces between brackets and the frames were modelled as bonded connections.

Model Details

The model parameters determine the quality of the elements that the assembly is split into, these elements determine the accuracy of the simulation.

- Nodes: 524179
- Elements: 253783
- Degrees of Freedom: 1592097
- Type of Analysis: Linear Static

The problem is characterized as linear static under the following assumptions:

- 1) Material behaviour is in the elastic region of deformation.
- 2) Relationship between the loads and displacement of the elements is linear.
- 3) The loads and constraints don't change in value or direction.

The load applied is incremented until yield point to ensure that the test stand is under the elastic region of deformation as the first assumption. This behaviour idealizes the behaviour that would be observed, since the force is only considered in the direction parallel to the line of symmetry of the test stand, assuming the force is equally distributed to both sides of the test stand.

Results

The goal of this analysis is to find the yield point load of the test stand and determine the FOS of the test stand at reasonably loads. Figure 6 represents the factor of safety as an inverse function of the applied load to the test stand. The trendline follows an inverse relationship with a R^2 value of 0.9939.

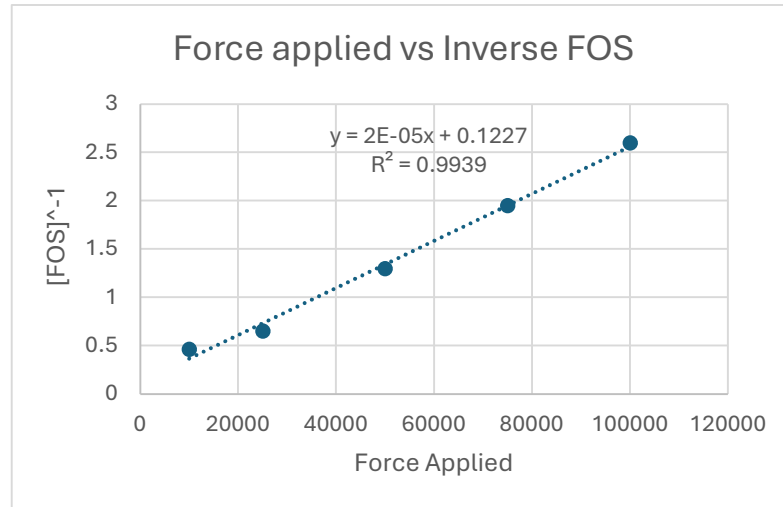


Figure 6: Applied Load vs Inverse Factor of Safety

The maximum allowable load is calculated to be 43865 N or 9861.2 lbF using the linear approximation between the applied force and the FOS. Figure 7 depict the Stress gradients of Ansys.

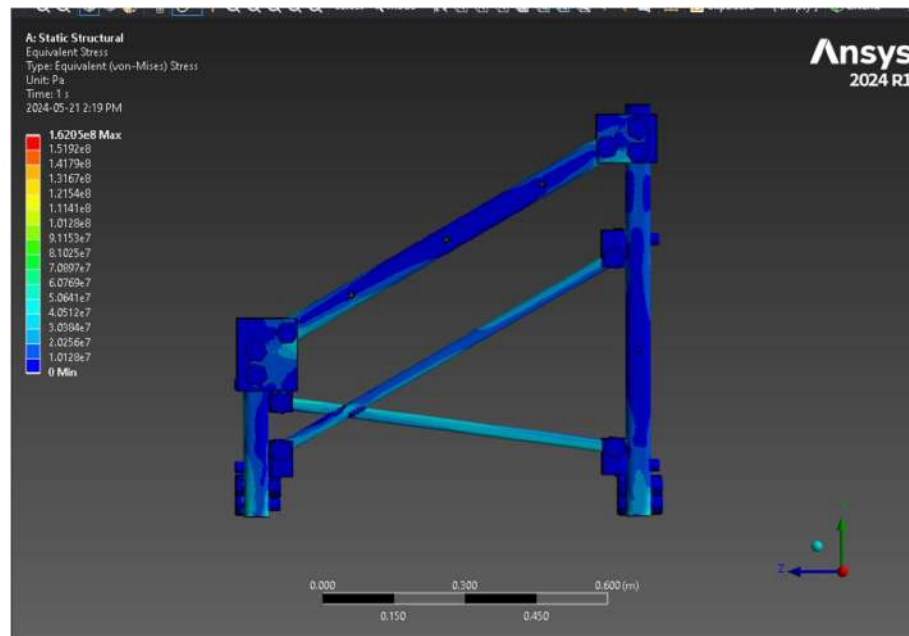


Figure 7: Equivalent Stress

Stress concentrations can be seen at the back of the test stand. The base of the test stand is anchored to crossbeams on the shipping container which allows for the force from the engine to be directly transmitted towards the y-axis of the shipping container.

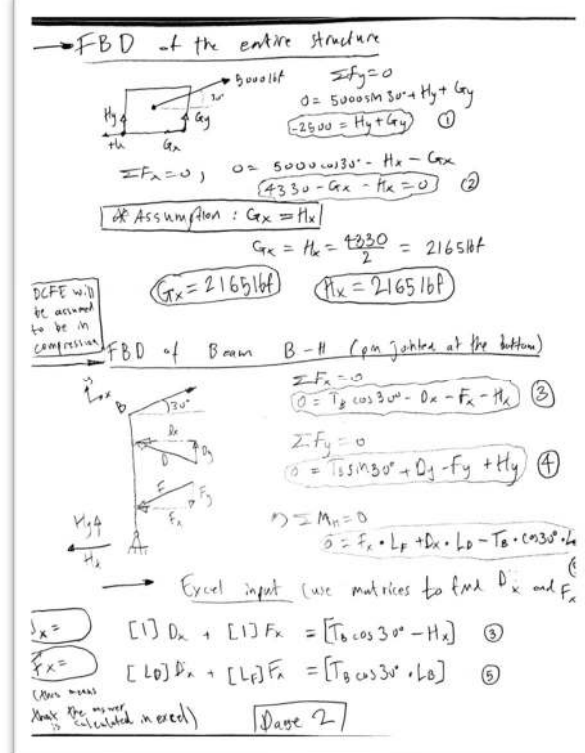
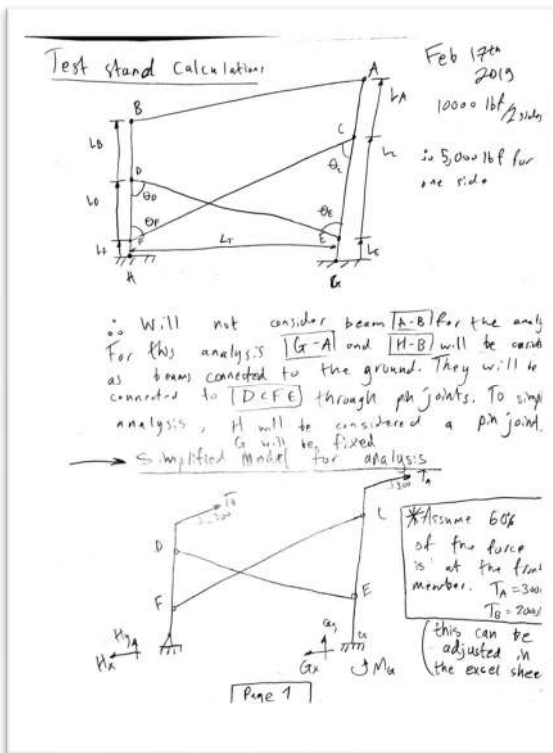
Conclusion

The results of FEA show that as designed the structural integrity of the test stand is maintained with a safety factor of 3.9 at an operating thrust of 1500 lbf or 6672.33 N. The maximum load of until yield of the test stand is 43865 N or 9861.2 lbf.

References

- [1] Ansys, [Online]. Available: <https://www.ansys.com/simulation-topics/what-is-finite-element-analysis>. [Accessed 18 05 2024].
- [2] "MatWeb Material Property Data," [Online]. Available: <https://www.matweb.com/search/DataSheet.aspx?MatGUID=10b74ebc27344380ab16b1b69f1cffbb>. [Accessed 20 05 2024].
- [3] E. Burlen, "Hex Head Bolt – a Basic Understanding," Tanner, 14 07 2022. [Online]. Available: <https://resources.tannerbolt.com/hex-head-bolt-a-basic-understanding/#:~:text=Hex%20bolts%2C%20also%20known%20as,grade%205%20zinc%20plated%20steel..> [Accessed 20 05 2024].
- [4] "304 Stainless Steel," MatWeb, [Online]. Available: <https://www.matweb.com/search/datasheet.aspx?MatGUID=abc4415b0f8b490387e3c922237098da>. [Accessed 20 05 2024].

Appendix A



→ Excel input (use matrices to find C_y and E_y)

$$[\tan \theta_E] C_y + [\tan \theta_E] E_y = [G_x - T_A \cos 30^\circ]$$

$$\begin{bmatrix} 1 \\ C_y \end{bmatrix} + \begin{bmatrix} -1 \\ E_y \end{bmatrix} = \begin{bmatrix} -G_x - T_A \sin 30^\circ \end{bmatrix}$$

$$\textcircled{1} C_x = C_y \tan \theta_E \quad \textcircled{2} E_x = E_y \tan \theta_E$$

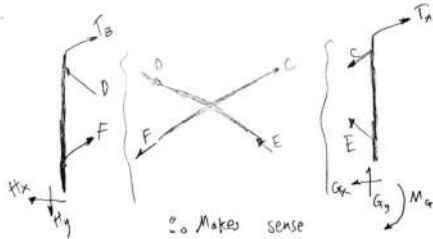
$$C_x = \quad E_x =$$

→ Find moment @ G

$$M_G = E_x \cdot L_E + C_x \cdot L_C + T_A \cdot \cos 30^\circ \cdot L_A \quad \textcircled{10}$$

$$M_G =$$

→ Model with oriented force directions



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*Assumption, D and F are single force members

$$\tan \theta_F = \frac{F_x}{F_y}, \quad F_y = \frac{F_x}{\tan \theta_F} \quad \textcircled{6} \quad \tan \theta_D = \frac{D_x}{D_y} \quad D_y = \frac{D_x}{\tan \theta}$$

$$F_y = \quad D_y = \quad |F| = \quad |D| = \quad (\text{all on excel sheet})$$

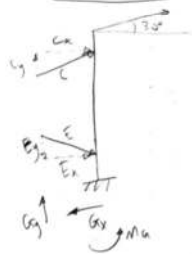
→ Find H_y and G_y

$$H_y = -T_A \sin 30^\circ - D_y + F_y \quad \textcircled{4}$$

$$G_y = -H_y - T_A \sin 30^\circ \quad \textcircled{1}$$

$$G_y =$$

→ FBD of A-G



$$\sum F_x = 0 \quad 0 = T_A \cos 30^\circ + C_x + E_x - G_x \quad \textcircled{8}$$

$$\sum F_y = 0 \quad 0 = T_A \sin 30^\circ + C_y - E_y + G_y \quad \textcircled{9}$$

$$\sum M_G = 0 \quad 0 = M_G - E_x \cdot L_E - C_x \cdot L_C - T_A \cdot \cos 30^\circ \cdot L_A \quad \textcircled{10}$$

Change $\textcircled{8}$

$$0 = T_A \cos 30^\circ + C_y \tan \theta_C + E_y \tan \theta_E - G_x$$

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